Tutorial title: Modern Methods in Associative Memory

Conference and year: International Conference on Machine Learning (ICML) 2025

Link: https://icml.cc/virtual/2025/40004

Description: The "Modern Methods in Associative Memory" tutorial in ICML 2025 explores how deep neural nets can be reformulated as energy-based dynamical systems. The main idea of the workshop is to follow the original, historical reformulation of classical Hopefield networks to dense Associative memories and apply the same process to modern neural nets, which results in the creation of the proposed energy transformer. I chose this tutorial because I do research that involves different neural network architectures (Transformers, CNNs, SSMs), especially for sequence modeling tasks, and I thought the energy-minimization perspective in this tutorial was interesting and could help me expand my research ideas.